

Max Mobley

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EXPERIENCE

Apple - Software Engineering Intern

May 2026 - Aug 2026 | Cupertino, CA

- Worked collaboratively across Apple Music Engineering to build new user-facing feature with low-level iOS integration
- Analyzed user access patterns to build custom on-device caching architecture for new sharing API endpoints

WPI Formula SAE Electric - Project Manager, Lead Driver

Aug. 2023 - Present | Worcester, MA

- Lead a team of 15 student engineers in designing an award-winning, ground-up electric race car under a strict ruleset
- Built digital twin and HITL testing pipelines to prevent firmware regressions and validate safety-critical behavior
- Built radio telemetry system, including PCB design (KiCad), PCB assembly, and embedded programming with STM32
- Created full-stack telemetry pipeline to parse and visualize 5,000+ events per second with near-zero end-to-end latency

Innovative Wireless Technologies - Systems Engineering Intern

May 2025 - Aug. 2025 | Lynchburg, VA

- Built and iterated on end-of-production test fixtures and comprehensive testing software for complex hardware
- Reduced net test time of existing fixtures by over 70% through software optimization and mechanical improvements

WPI Academic & Research Computing - Server Technician

Sep. 2022 - May 2026 | Worcester, MA

- Serviced and upgraded WPI academic research cluster, including Dell EMC, Arista, and Juniper hardware systems
- Built software tools to assist students in using cluster resources, such as an SSO-enabled RDP authentication proxy

WPI CollabLab Makerspace - President

Aug. 2022 - Present | Worcester, MA

- Assisted students with personal projects and taught them to use tools, CAD, 3D printers, programming, and electronics
- Maintained tools, machines, and a physical lab space while also organizing events to foster a community of makers

NerdSpeak Technologies - Software Engineer

June 2021 - May 2022 | Remote

- Built software systems which helped over 100,000 users purchase hard-to-find tech items during electronics shortages
- Designed and optimized a high-throughput, distributed software architecture to achieve lowest-latency web scraping

PROJECTS

Autonomous Battlebot: Built robot and supporting software for an AI fighting robot with numerous sensors, PCBs, and custom ML models, which altogether powered a custom control pipeline with low latency and complex awareness

Digital Website Archival: Built high throughput web scrapers and reverse engineered web APIs to save decaying websites while volunteering with the Internet Archive to ensure multiple copies are saved and made available to the public

Home Server Infrastructure: Created and maintained fully redundant networking and compute in a home environment and architected multi-hundred-terabyte storage systems to serve as a data warehouse for website archival operations

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, Bash, Golang

Frameworks & Technologies: S3, SQLite, Ceph, ElasticSearch, git, Unix, Linux, Proxmox, Cisco, Jira, Grafana, VSCode, Docker (Swarm), Solidworks, Flask, KiCAD, STM32, Postgres, NGINX, OpenCV, mitmproxy, Selenium, I2C, SPI, CAN

Concepts: Networking, Distributed Systems, Backend & REST API Development, CI/CD, Reverse Engineering, High Throughput Web Scraping, PCBA, Database Programming, Computer Vision, Oscilloscope, Logic Analyzer, DevOps

Soft Skills: Remote & In-Person Experience, Leadership, Collaboration, Self-Teaching, Public Speaking, Technical Writing

EDUCATION

Worcester Polytechnic Institute

B.S. Computer Science | GPA: 3.5

Class of 2026

Worcester, MA

Coursework: Operating Systems, Machine Assembly, Algorithms, Database Systems, Computer Networking, Software Engineering, Intro to Computer Engineering, Electrical Systems, Sensors & Circuits, Intro to Robotics Engineering

Research (Lyon, FR - 2024): Conducted interviews, research, and wrote a 130-page report on Lyon's public transit system